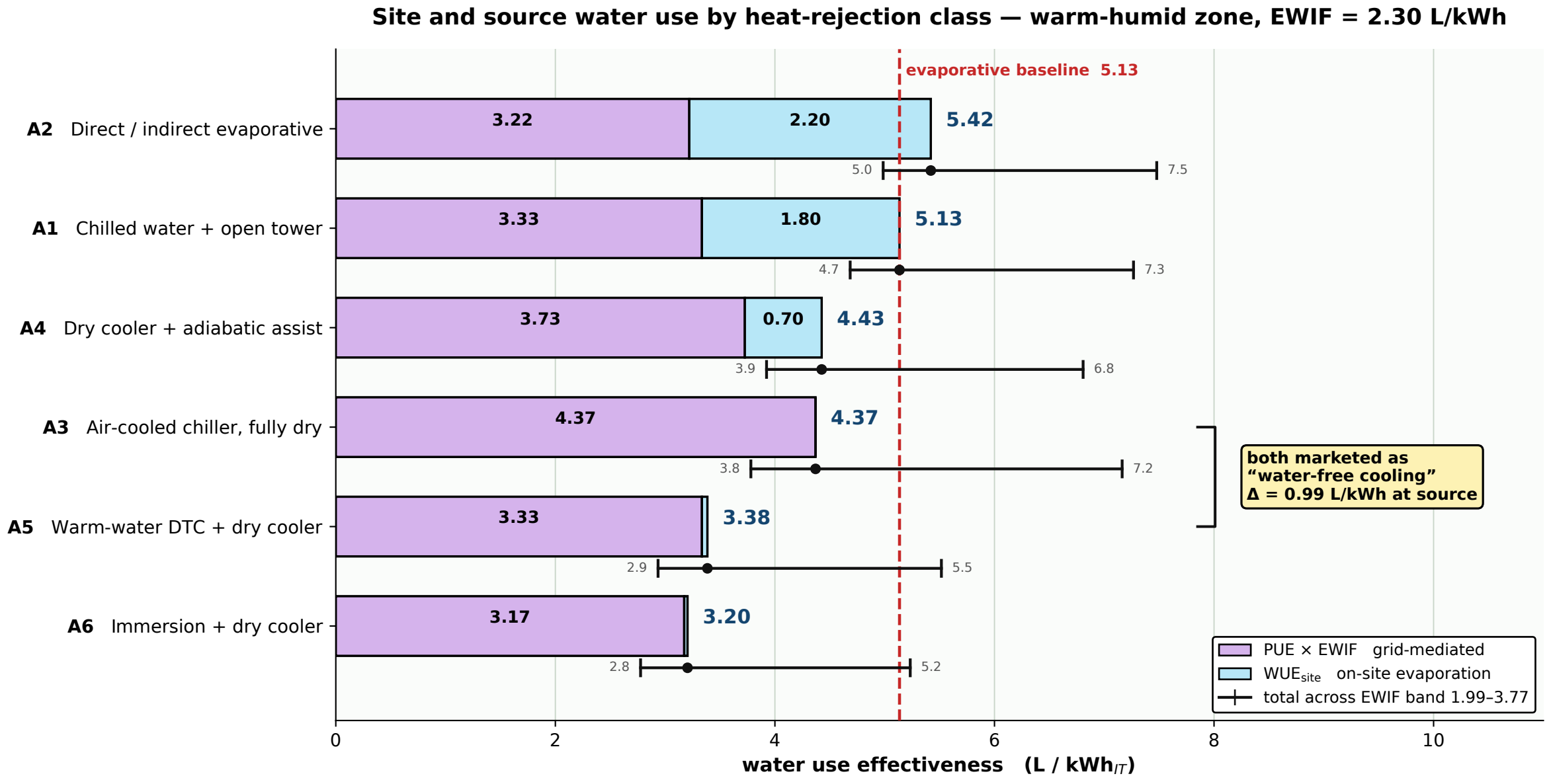
Site and source water use by heat-rejection class — warm-humid zone, EWIF = 2.30 L/kWh
evaporative baseline 5.13
A2 Direct / indirect evaporative
3.22
2.20
5.42
5.0
7.5
A1 Chilled water + open tower
3.33
1.80
5.13
4.7
7.3
A4 Dry cooler + adiabatic assist
3.73
0.70
4.43
3.9
6.8
A3 Air-cooled chiller, fully dry
4.37
4.37
3.8
7.2
A5 Warm-water DTC + dry cooler
3.33
3.38
2.9
5.5
A6 Immersion + dry cooler
3.17
3.20
2.8
5.2
both marketed as
“water-free cooling”
Δ = 0.99 L/kWh at source
PUE × EWIF grid-mediated
WUE$_{site}$ on-site evaporation
total across EWIF band 1.99–3.77
0
2
4
6
8
10
water use effectiveness (L / kWh$_{IT}$)